SEC 1 · LEC 3 · TYPES OF INTERPLANETARY SPACECRAFT

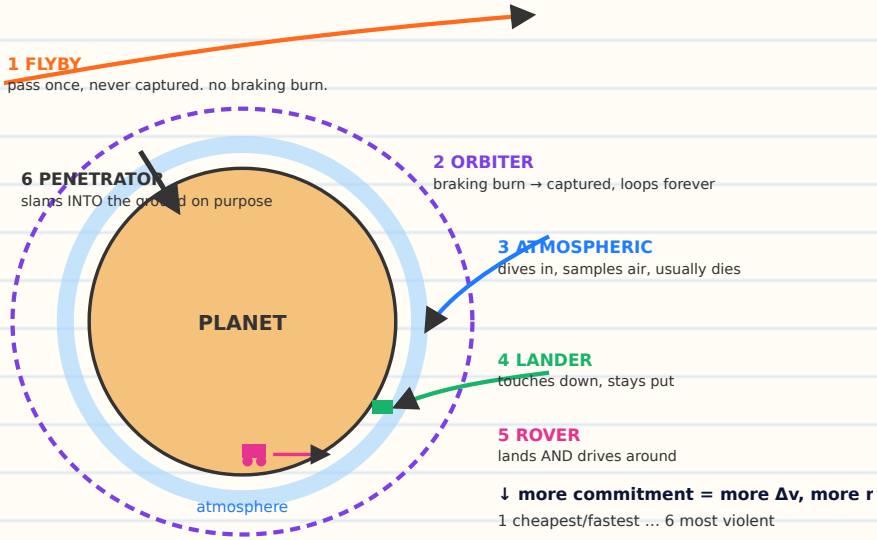
The slide is a ladder. Draw the ladder.

Eight bullet points on the slide. Six of them are the same idea at increasing commitment: how close do you get to the planet? The other two aren't about a planet at all.

🖉 LEFT PAGE · DRAW

🚀 SEC 1 / LEC 3 — TYPES OF SPACECRAFT

date · pg __



🔭 THE OTHER TWO aren't about a planet

7. OBSERVATORY — a telescope parked in space (Hubble, JWST). The target is the sky, not a body.
8. COMMUNICATIONS — a relay so the ones above can phone home (Mars orbiters relay for rovers; China's Queqiao relays for the Moon's far side).

❓ RIGHT PAGE · CORNELL

CUES = questions | **NOTES = keywords**

Rank the 6 "go to a planet" types from least to most committed.

Which type needs NO braking burn? Why is that a big deal?

Why ever fly by if you could orbit?

Atmospheric probe vs lander: what's the difference?

Which 2 types aren't aimed at a planet at all?

? did any penetrator mission actually work

FLYBY — pass once · no capture burn · can chain planets (Voyager grand tour, New Horizons → Pluto)
ORBITER — brake into orbit · long stay · maps (MRO, Cassini, Juno)
ATMOSPHERIC — parachute/balloon into the air · short life (Galileo probe → Jupiter, Huygens → Titan)
LANDER — survives touchdown · fixed spot (Philae, Viking, InSight)
ROVER — lander + wheels (Sojourner, Spirit/Opportunity, Curiosity, Perseverance)
PENETRATOR — impact on purpose · buries sensors or kicks up material (Deep Impact → comet, LCROSS → Moon)
OBSERVATORY — telescope platform (Hubble, JWST)
COMMS — relay for the others

ladder: closer = more Δv to slow down = heavier = riskier = more data
flyby wins on: cost, speed, multi-target. loses on: time at target (hours)
one craft can be 2 types: Cassini (orbiter) carried Huygens (atmospheric + lander)

🧠 SUMMARY — video closed, from memory (5 lines)

1. _____
2. _____
3. _____
4. _____
5. _____

WHY THIS SLIDE MATTERS

It's not a vocabulary list. It's a cost menu.

The instructor is setting up the whole course here. Every later section (propulsion, comms, power) is "what does it cost to climb one more rung of this ladder?"

🪜 The ladder, in plain English

- **Flyby:** you drive past the house at highway speed and take photos out the window. Cheap, fast, and you can hit several houses on one trip. But you get a few hours, once.
- **Orbiter:** you pull into the driveway and stay. That means **braking**, and braking costs fuel you had to carry all the way there. In return: months or years of looking.
- **Atmospheric:** you throw a sensor out the window as you pass. It parachutes down, reports until it's crushed or out of battery. Short life, but it's the only way to taste the air.
- **Lander:** you actually get out. Surviving the landing is the hard part (entry, descent, landing — "seven minutes of terror" on Mars).
- **Rover:** a lander that can walk around. Everything a lander needs, plus wheels, plus way more power and comms.
- **Penetrator:** you throw a dart into the lawn to see what's under the grass. Violent, cheap-ish, high failure rate.

⚡ The one sentence to remember: every rung down the ladder costs more fuel to slow down, more mass, more risk — and buys more science.

🔎 Fact checks on the slide

- The caption said a flyby goes "around" the planet. Sharpen that: it goes **past** the planet, bending its path but never getting captured. "Around" is what an orbiter does.
- Philae (the lander on the slide) is a good example of why landers are hard: it bounced twice on the comet and ended up in a shadow with no solar power. Right type, rough landing.
- The rover pictured is actually a Mars Exploration Rover (Spirit/Opportunity design), not Curiosity. Minor, but notice it: slides have errors too. Verify, don't just absorb.

📒 Question bank cards

- Rank: flyby, orbiter, atmospheric, lander, rover, penetrator — by commitment. What rises with each step?
- Which spacecraft type needs no braking burn at the target?
- Give one real mission for each of the 8 types.
- Why did Voyager fly by four planets instead of orbiting one?
- Observatory and communications spacecraft: what's their target?

✏️ In your notebook, right now

Pause. On your left page, draw a circle (planet) and six lines showing each type's path, numbered 1–6. Then write the ladder sentence under it. That's the whole lecture in one drawing. Then press play.

SEC 1 · LEC 3 · VOCABULARY

Types of spacecraft, in plain English

Every word used on the previous pages, including the ones I introduced in the notes column. Read once; on the +1 day review, explain three of them out loud without looking.

Interplanetary — between planets. An interplanetary spacecraft leaves Earth's neighborhood entirely, as opposed to a satellite that stays in Earth orbit.

Orbiter — a spacecraft that fires its engine to slow down at the planet so gravity captures it into orbit. *Pulling into the driveway and staying.*

Lander — a craft that survives touchdown and sits in one place doing science (Viking, InSight, Philae).

Penetrator — a craft designed to slam into the surface on purpose, either to bury sensors or to kick up material for another craft to analyze (Deep Impact hit a comet; LCROSS hit the Moon).

Communications spacecraft — a relay whose job is passing messages between other spacecraft and Earth. Mars orbiters relay for the rovers; China's Queqiao relays for the far side of the Moon.

Braking burn / capture burn — firing the engine against your direction of travel when you arrive, to slow down enough for the planet's gravity to hold you. A flyby skips this; an orbiter can't.

Gravity assist — swinging close past a planet to steal a bit of its orbital speed, for free. New Horizons used Jupiter this way. *Grabbing a moving merry-go-round to fling yourself faster.*

Commitment ladder — my phrase, not the instructor's: flyby → orbiter → atmospheric → lander → rover → penetrator, ordered by how much fuel, mass, and risk each needs, and how much science it returns.

Kuiper Belt — the ring of icy bodies beyond Neptune. Pluto is the biggest known member. New Horizons kept flying into it after Pluto.

Voyager — two 1977 NASA flyby craft that visited Jupiter, Saturn, Uranus and Neptune on one trip, then left the solar system. The classic example of why a flyby can do more targets.

Cassini / Huygens — Cassini = NASA Saturn orbiter (2004–2017). Huygens = the European probe it carried, which parachuted onto Titan. One mission, two types.

InSight / Philae / Rosetta — InSight = NASA Mars lander (2018) that listened for marsquakes. Rosetta = European comet orbiter; Philae = its lander, which bounced into a shadow and lost power.

Queqiao — China's relay satellite parked beyond the Moon so the Chang'e-4 lander on the far side could talk to Earth.

Flyby — a spacecraft that passes a planet once at high speed and keeps going. It never slows down enough to be caught. *Driving past a house and photographing it out the window.*

Atmospheric probe — a small craft that dives into a planet's air on purpose, usually on a parachute or balloon, and reports until it's crushed or runs out of power (Galileo probe at Jupiter, Huygens at Titan).

Rover — a lander with wheels: it can drive to new spots (Sojourner, Spirit, Opportunity, Curiosity, Perseverance).

Observatory spacecraft — a telescope platform whose target is the sky, not a planet (Hubble, JWST, Kepler).

Δv (delta-v) — "change in velocity": how much speeding up or slowing down a mission needs, in km/s. It's the real currency of spaceflight, because every bit of Δv costs fuel you had to carry. *Read it as "fuel budget."*

Prograde / retrograde — prograde = firing so you speed up along your direction of motion; retrograde = firing to slow down. *Gas pedal vs brake.*

EDL — Entry, Descent, Landing: the sequence a lander or rover goes through from hitting the top of the atmosphere to touching down. On Mars it's ~7 minutes and fully automatic. *"Seven minutes of terror."*

C3 (launch energy) — a number for how hard you have to throw a spacecraft to leave Earth and reach a target, in km²/s². Bigger C3 = bigger rocket. Earth escape is C3 = 0; Mars needs ~10–15; Pluto direct needed far more.

Comet — a chunk of ice and dust that grows a tail when it nears the Sun. Rosetta orbited one and dropped the Philae lander onto it.

New Horizons — the 2006 NASA flyby craft that reached Pluto in 2015 (9.5 years). Launched too fast to ever stop there.

Juno / MRO — Juno = NASA Jupiter orbiter. MRO = Mars Reconnaissance Orbiter, which also relays data for rovers on the ground.

Deep Impact / LCROSS — Deep Impact (2005) fired an impactor into comet Tempel 1. LCROSS (2009) crashed a rocket stage into a lunar crater to look for water in the spray. Both are penetrator-style missions.